

ATHARVA BHORPE

Machine Learning & Robotics | Deep Learning · LLMs · Computer Vision · ROS 2

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CORE COMPETENCIES

Machine Learning & AI: Python, Deep Learning, PyTorch, LLMs, Computer Vision

Robotics Frameworks: ROS 2, MoveIt 2, Gazebo, LeRobot

Robotics Concepts: Kinematics, Dynamics, Motion & Path Planning

Tools & Platforms: Linux, Git, NumPy, Fusion 360

Languages: English (C1), German (A1, A2 in progress)

EDUCATION

MSc, Robotic Systems Engineering

Sep 2024 – Present

RWTH Aachen University, Germany

B.E., Mechanical Engineering

Aug 2018 – May 2022

Pimpri Chinchwad College of Engineering, India

PROFESSIONAL EXPERIENCE

Application Engineer

Jul 2022 – Aug 2024

Wipro PARI, Pune, India

- Developed pre-sales technical concepts and commercial offers for global vehicle and EV-battery automation projects supporting VW-Skoda, VinFast and SKF across Vietnam, the USA, and India.
- Specified automation hardware across 6 vendors by mapping customer requirements to Cognex vision, ABB/Fanuc robots, Siemens/Bosch Rexroth controls, and Atlas Copco tooling.
- Supported automation proposals through component selection, procurement planning and solution costing.

Robotics Design Intern

May 2021 – Aug 2021

Ollivander Robotics, Pune, India

- Built and tested the ROS 1 mobile robot prototype with autonomous navigation capabilities.
- Designed and built an autonomous mobile robot chassis using Fusion 360, progressing from CAD to a working prototype.

PROJECTS

Tiny Recursive Model for Path Planning

May 2026 – June 2026

PyTorch, Deep Learning, Transformers

- Achieved 100% path-success at optimal (1.0 ratio) route length on 26×26 occupancy grids by building a 1.2M-parameter recursive transformer that learns the per-cell action policy (N/S/E/W/stay).
- Matched the effective depth of a far deeper network within a 2-layer ($\sim 1.2M$) parameter budget and under 2 GB VRAM by exploiting weight-shared recursion, keeping the model edge- and robotics-deployable.
- Recovered out-of-distribution success from 0.52 to 0.75 on 40×40 grids ($1.5 \times$ the trained size) by applying test-time recursion scaling, validated by benchmarking optimality and latency against A* and parameter-matched CNN baselines.

AI for Industry Challenge: Autonomous Insertion (Intrinsic)

Mar 2026 – May 2026

LeRobot, PyTorch, Diffusion Policy, Gazebo

- As part of a team, placed 35th in Intrinsic's global AI for Industry Challenge, training imitation-learning policies to insert fiber-optic plugs (SFP/SC) with a simulated UR5e arm in Gazebo.
- Tuned a Diffusion Policy (per-camera ResNet34 encoders, wider U-Net, extended action horizon) that outperformed an ACT baseline, scoring 86/100 on the SFP subtask.
- Improved robustness to distribution shift using DAGger data collection, and trained models on the RWTH HPC cluster.

Robotic Arm Control using IMU

Nov 2025 – Jan 2026

ROS 2, Python, Arduino, MoveIt

- Built a teleoperation system commanding a simulated Kinova arm in Cartesian space via MoveIt, running raw UDP on an ESP32.
- Fused IMU, rotary-encoder, and IR-sensor inputs and streamed twist commands over WiFi: IMU for X-Y axes, encoder for Z, IR for gripper control.

Material Handling AMR — Bachelor's Thesis

Jan 2022 – Apr 2022

Python, ROS 1 Navigation Stack, OpenCV

- Built a mobile robot transporting a 5 kg payload at 0.3 m/s with hand-signal recognition, for under €300.
- Configured ROS 1 with the Navigation Stack: Raspberry Pi for perception, localization, mapping, and navigation; ESP32 for low-level motor and sensor control.
- Selected for an Indian National Robotics Conference and published in ACM as an International Conference Proceedings Series (ICPS) paper.

PUBLICATIONS

- “Autonomous Mobile Robot Base: A Cost-Effective Solution for Research & Education” — ACM ICPS, AIR '23.
- “Autonomous Navigation on an Omni-Wheeled Robot Using 2D LiDAR & ROS” — LNNS, Springer (BigDML 2021).

CERTIFICATIONS

- ROS 2 for Beginners
- Learn ROS2 as a ROS1 Developer and Migrate Your ROS Projects